

We saw in 11.9 that we can write a power series representation of some functions. However, we are very limited on the amount of functions we can do this for.

In this section we will find out how to make series representations for more functions like  $\sin(x)$ ,  $\cos(x)$ ,  $e^x$ .

Let's start by assuming  $f$  has a power series expansion, such functions are called analytic. That is

$$f(x) = C_0 + C_1(x-a) + C_2(x-a)^2 + C_3(x-a)^3 + \dots \quad |x-a| < R$$

Let's try to find the values of  $C_n$ .

$$\begin{aligned} \text{Notice } f(a) &= C_0 + C_1(a-a) + C_2(a-a)^2 + \dots \\ &= C_0 \end{aligned}$$

If we differentiate

$$f'(x) = C_1 + 2C_2(x-a) + 3C_3(x-a)^2 + 4C_4(x-a)^3 + \dots$$

$$f''(x) = 2C_2 + 2 \cdot 3C_3(x-a) + 3 \cdot 4C_4(x-a)^2 + 4 \cdot 5C_5(x-a)^3 + \dots$$

$$f'''(x) = 2 \cdot 3C_3 + 2 \cdot 3 \cdot 4(x-a) + 3 \cdot 4 \cdot 5(x-a)^2 + 4 \cdot 5 \cdot 6(x-a)^3 + \dots$$

Continuing

$$f^{(n)}(x) = n!C_n + (n+1)!(x-a) + (n+2)!(x-a)^2 + \dots$$

↳  $n^{\text{th}}$  derivative

Evaluating at  $a$

$$f^{(n)}(a) = n!C_n \rightarrow \frac{f^{(n)}(a)}{n!} = C_n$$

Recall  $f^{(0)} = f$  and  $0! = 1$

Thm:

If  $f$  has a power series expansion at  $a$ ,  
i.e.  $f(x) = \sum_{n=0}^{\infty} C_n(x-a)^n$   $|x-a| < R$

Then the coefficients  $C_n$  are given by

$$C_n = \frac{f^{(n)}(a)}{n!}$$

Plugging back in, this means

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$$

This is called the Taylor Series expansion of  $f$  at  $x=a$  (or centered at  $a$ )

In the special case that  $x=0$ , then

$$f(x) = f(c) + f'(c)x + \frac{f''(c)}{2}x^2 + \frac{f'''(c)}{3!}x^3 + \dots$$

is called a Maclaurin Series.

Note:

1) The theorem starts with "if the function has a power series representation"  
Not all functions can be written as a Taylor Series

2) The theorem explains that these expansions are unique. So any other power series expansion is the same as the Taylor Series.

Ex: we know that

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad \text{from before.}$$

Let's make sure the Taylor Series agrees.

|                                       |                   |
|---------------------------------------|-------------------|
| $f(x) = \frac{1}{1-x}$                | $f(0) = 1$        |
| $f'(x) = \frac{1}{(1-x)^2}$           | $f'(0) = 1$       |
| $f''(x) = \frac{2}{(1-x)^3}$          | $f''(0) = 2!$     |
| $f'''(x) = \frac{2 \cdot 3}{(1-x)^4}$ | $f'''(0) = 3!$    |
| $\vdots$                              | $\vdots$          |
| $f^{(n)}(x) = \frac{n!}{(1-x)^{n+1}}$ | $f^{(n)}(0) = n!$ |

$$\text{So, } C_n = \frac{f^{(n)}(0)}{n!} = \frac{n!}{n!} = 1$$

$$\text{So } \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad \text{like before}$$

Ex: Maclaurin Series for  $f(x) = e^x$ .

|                |              |
|----------------|--------------|
| $f(x) = e^x$   | $f(0) = 1$   |
| $f'(x) = e^x$  | $f'(0) = 1$  |
| $f''(x) = e^x$ | $f''(0) = 1$ |
| $\vdots$       | $\vdots$     |

$$\text{So, } e^x = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + f'(0)x + f''(0)x^2 + \dots$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

From the ratio test

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{x^{n+1}}{(n+1)!}}{\frac{x^n}{n!}} \right| = \lim_{n \rightarrow \infty} \frac{x}{n+1} = 0 \text{ for all } x$$

$\therefore$  this series has radius of convergence  $\infty$ .

Note: This example shows that that if  $e^x$  has a power series expansion at 0, it would be  $\sum \frac{x^n}{n!}$ .

How do we know when a function has a power series expansion?

Take

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

denote

$$T_n(x) = \sum_{i=0}^n \frac{f^{(i)}(a)}{i!} (x-a)^i \quad \left( \begin{array}{l} \text{similar to } n^{\text{th}} \\ \text{partial sum} \end{array} \right)$$

$T_n$  is called the  $n^{\text{th}}$  degree Taylor Polynomial  
of  $f$  at  $a$

In the case of  $e^x$

$$T_0 = 1, \quad T_1 = 1+x, \quad T_2 = 1+x+\frac{x^2}{2}, \\ T_3 = 1+x+\frac{x^2}{2}+\frac{x^3}{6}$$

like with partial sums

$$f(x) = \lim_{n \rightarrow \infty} T_n(x)$$

For any  $n$ , define  $R_n(x) = f(x) - T_n(x)$

$R_n(x)$  is called the  $n^{\text{th}}$  Taylor remainder

Since  $f(x) = T_n(x) + R_n(x)$ ,

if we show  $\lim_{n \rightarrow \infty} T_n(x) = 0$ , then  $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} x^n$

Thm:

If  $f(x) = T_n(x) + R_n(x)$ , where  $T_n$  is the  $n^{\text{th}}$  Taylor Polynomial of  $f$  at  $a$ , and if

$$\lim_{n \rightarrow \infty} R_n(x) = 0$$

For  $|x-a| < R$ , then  $f$  is equal to the sum of its Taylor Series on its Interval of Convergence.

In Practice, to show  $\lim_{n \rightarrow \infty} R_n(x) = 0$

we use the following theorem

Thm:

If  $|f^{(n+1)}(x)| \leq M$  for  $|x-a| \leq d$ , then  $R_n(x)$  satisfies the inequality

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1} \quad \text{for } |x-a| \leq d$$

Note:  $\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$  for all  $x$ .

Ex: Show  $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$  (Maclaurin Series)

If  $f(x) = e^x$ ,  $f^{(n)}(x) = e^x$  for all  $n$ .

say  $d > 0$  s.t.  $|x| \leq d$ , then  $|f^{(n+1)}(x)| = e^x \leq e^d$

So,

$$|R_n(x)| \leq \frac{e^d}{(n+1)!} |x|^{n+1} \quad \text{for } |x| \leq d$$

Note  $0 \leq |R_n(x)|$ , and

$$\lim_{n \rightarrow \infty} \frac{e^d}{(n+1)!} |x|^{n+1} = e^d \lim_{n \rightarrow \infty} \frac{|x|^{n+1}}{(n+1)!} = e^d \cdot 0 = 0$$

So  $\lim_{n \rightarrow \infty} R_n(x) = 0$  and  $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$  for all  $x$ .



Ex:  $e^x$  centered at  $a=2$

$$f(x) = e^x$$

$$f'(x) = e^x$$

$$\vdots$$

$$f(2) = e^2$$

$$f'(2) = e^2$$

$$\vdots$$

$$f^{(n)}(2) = e^2$$

$$e^x = \sum_{n=0}^{\infty} \frac{e^2}{n!} (x-2)^n$$

We will now find Taylor Series expansions for  $\sin(x)$ , and  $\cos(x)$

Ex:  $f(x) = \sin(x)$  at  $x=0$

$$\sin(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \dots$$

$$= 0 + x + 0 - \frac{x^3}{3!} + 0 + \frac{x^5}{5!}$$

$$= \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$$

$\leadsto$  Radius of convergence is  $\infty$

$$f(x) = \sin(x)$$

$$f'(x) = \cos(x)$$

$$f''(x) = -\sin(x)$$

$$f'''(x) = -\cos(x)$$

$$f^{(4)}(x) = \sin(x)$$

$$f(0) = 0$$

$$f'(0) = 1$$

$$f''(0) = 0$$

$$f'''(0) = -1$$

$$f^{(4)}(0) = 0$$

Ex:  $\cos(x)$  at  $x=0$

$$\cos(x) = 1 + 0x - \frac{x^2}{2!} + 0x^3 + \frac{x^4}{4!} - \dots$$

$$= \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

Radius of convergence  
 $R = \infty$

|       | $f^{(n)}(x)$ | $f^{(n)}(0)$ |
|-------|--------------|--------------|
| $n=0$ | $\cos(x)$    | 1            |
| $n=1$ | $-\sin(x)$   | 0            |
| $n=2$ | $-\cos(x)$   | -1           |
| $n=3$ | $\sin(x)$    | 0            |
| $n=4$ | $\cos(x)$    | 1            |
|       | $\vdots$     | $\vdots$     |

Ex:  $f(x) = (1+x)^k$  for  $k > 0$  about  $x=0$

|          | $f^{(n)}(x)$                    | $f^{(n)}(0)$                             |
|----------|---------------------------------|------------------------------------------|
| $n=0$    | $(1+x)^k$                       | 1                                        |
| $n=1$    | $k(1+x)^{k-1}$                  | $k$                                      |
| $n=2$    | $k(k-1)(1+x)^{k-2}$             | $k(k-1)$                                 |
| $n=3$    | $k(k-1)(k-2)(1+x)^{k-3}$        | $k(k-1)(k-2)$                            |
| $\vdots$ | $\vdots$                        | $\vdots$                                 |
| $n$      | $k(k-1)\dots(k-n+1)(1+x)^{k-n}$ | $k(k-1)\dots(k-n+1) = \frac{k!}{(n-k)!}$ |

$$\text{So } (1+x)^k = \sum_{n=0}^{\infty} \frac{k!}{(n-k)!n!} x^n = \sum_{n=0}^{\infty} \binom{k}{n} x^n \quad \text{Converges for } |x| < 1$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \dots$$

$$R=1$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$R=\infty$$

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$R=\infty$$

$$\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$R=\infty$$

$$\tan^{-1}(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

$$R=1$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$R=1$$

$$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n = 1 + kx + \frac{k(k-1)}{2!} x^2 + \dots$$

$$R=1$$

We can use these series to make new series

Ex:  $f(x) = x \cos x$

$$\cos x = \sum (-1)^n \frac{x^{2n}}{(2n)!}$$

$$x \cos(x) = \sum (-1)^n \frac{x^{2n+1}}{(2n)!}$$

$f(x) = \ln(1+3x^2)$

$$\ln(1+x) = \sum (-1)^{n-1} \frac{x^n}{n}$$

$$\ln(1+3x^2) = \sum (-1)^{n-1} \frac{(3x^2)^n}{n}$$

Ex: Evaluate  $\int e^{x^2} dx$

$$e^x = \sum \frac{x^n}{n!}$$

$$e^{x^2} = \sum \frac{x^{2n}}{n!}$$

Converges for all  $x$

$$\int e^{x^2} dx = C + \sum \int \frac{x^{2n}}{n!} dx = C + \sum \frac{x^{2n+1}}{(2n+1)n!}$$

Ex:

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$$

using Taylor Series

$$= \lim_{x \rightarrow 0} \frac{(1 + x + \frac{x^2}{2} + \dots) - 1 - x}{x^2} = \lim_{x \rightarrow 0} \frac{\frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!}}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{1}{2} + \frac{x}{3!} + \frac{x^2}{4!} + \dots$$

$$\boxed{= \frac{1}{2}}$$

Ex:

$$\frac{1}{1 \cdot 2} - \frac{1}{2 \cdot 2^2} + \frac{1}{3 \cdot 2^3} - \dots$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n 2^n} = \sum_{n=1}^{\infty} \frac{(\frac{1}{2})^n}{n} (-1)^{n+1} \rightarrow \ln(x+1) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} \text{ for } |x| < 1$$
$$= \ln\left(1 + \frac{1}{2}\right) = \ln\left(\frac{3}{2}\right)$$

Ex: Second degree Taylor polynomial for  $x^{\frac{1}{2}}$  at  $a=1$

| $n$ | $f^{(n)}(x)$                    | $f^{(n)}(1)$   |
|-----|---------------------------------|----------------|
| 0   | $x^{\frac{1}{2}}$               | 1              |
| 1   | $\frac{1}{2} x^{-\frac{1}{2}}$  | $\frac{1}{2}$  |
| 2   | $-\frac{1}{4} x^{-\frac{3}{2}}$ | $-\frac{1}{4}$ |

$$T_2(x) = 1 + \frac{1}{2}(x-1) - \frac{1}{8}(x-1)^2$$
$$= 1 + \frac{1}{2}x - \frac{1}{8}x^2$$

$$\int_{-\infty}^{\infty} \sqrt{x} \cos(x^2) dx$$

$$\cos(x) = \sum \frac{(-1)^n x^{2n}}{(2n)!}$$

$$\cos(x^2) = \sum \frac{(-1)^n x^{4n}}{(2n)!}$$

$$\sqrt{x} \cos(x^2) = \sum \frac{(-1)^n x^{4n+\frac{1}{2}}}{(2n)!}$$

$$\int \sqrt{x} \cos(x^2) dx = C + \sum \frac{(-1)^n x^{4n+\frac{3}{2}}}{(4n+\frac{3}{2})(2n)!}$$